

Systematic Momentum Portfolio with Diversified Selection and Risk-Based Allocation

Finesse $\times$ Citadel Portfolio Construction Challenge – Round 2

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Team Name - Lost Physicists

Initial Capital: Rs. 1,00,00,000 | Maximum Holdings: 10 | Backtest: 1 Jan 2021–31 Dec 2025

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Problem and Strategy Overview

The objective is to construct and manage a systematic equity portfolio from the permitted Nifty 100, Nifty Midcap 100 and Nifty Smallcap 100 universe, subject to a maximum of ten holdings and an initial virtual corpus of Rs. 1 crore. The strategy is evaluated over 2021–2025 with a transaction cost of 0.1% applied to every transaction.

The central idea is deliberately simple: **rank stocks by intermediate-term momentum, retain only stocks in a long-term positive trend, and construct a diversified portfolio through sector and correlation constraints**. Volatility is **not used as an alpha or stock-selection factor**; it is used only for basic eligibility screening and for risk-based capital allocation after stocks have been selected.

The methodology has four layers:

1. **Momentum ranking:** stocks are ranked solely by approximately 12-minus-1-month price momentum.
2. **Trend and data-quality filtering:** stocks must trade above their 250-day moving average and have sufficient return observations.
3. **Diversified selection:** sector concentration and pairwise correlation limits prevent the portfolio from becoming a collection of highly similar momentum bets.
4. **Risk-based allocation:** selected stocks are weighted inversely to their realized volatility.

This separation between *alpha generation* and *risk management* is intentional. Momentum determines *what* to own; diversification and inverse-volatility weighting determine *how* the selected positions are combined.

End-to-End Strategy Workflow

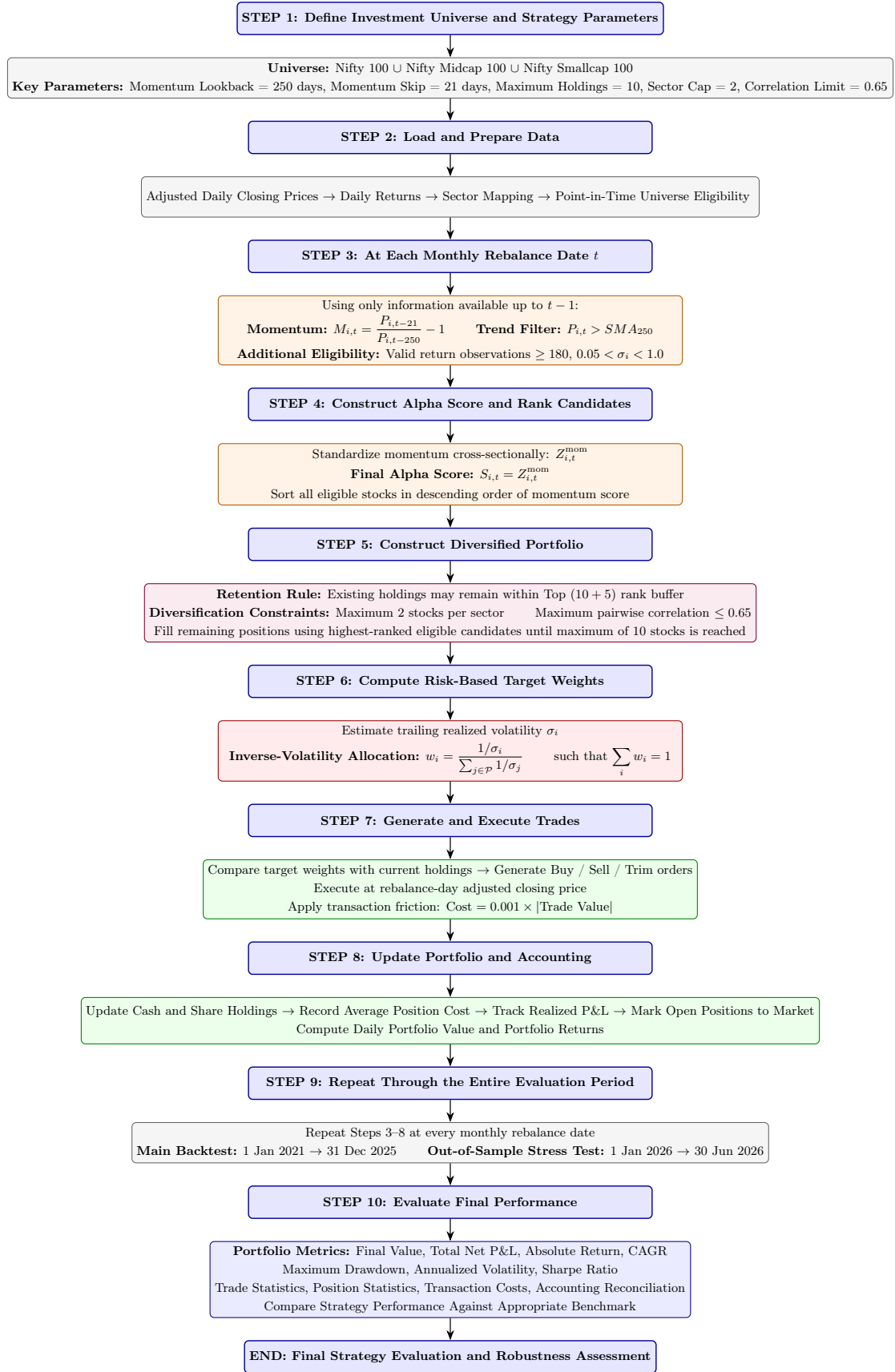


Figure 1: End-to-end workflow of the systematic momentum portfolio strategy. Momentum is used exclusively for alpha generation and stock ranking, whereas volatility is used subsequently for risk-based portfolio allocation.

3.1 Momentum signal

Let $P_{i,t}$ denote the adjusted closing price of stock i . The primary ranking signal uses a 250-trading-day lookback while skipping the most recent 21 trading days:

$$M_{i,t} = \frac{P_{i,t-21}}{P_{i,t-250}} - 1. \quad (1)$$

This approximately 12-minus-1-month construction targets intermediate-term price persistence while avoiding excessive sensitivity to the most recent month, where short-term reversals and temporary price shocks can dominate.

Momentum is standardized cross-sectionally among eligible candidates:

$$Z_{i,t}^{\text{mom}} = \frac{M_{i,t} - \mu_{M,t}}{\sigma_{M,t} + 10^{-6}}. \quad (2)$$

Because the volatility-selection weight is explicitly zero in the final model, the final ranking score is simply

$$S_{i,t} = Z_{i,t}^{\text{mom}}. \quad (3)$$

Thus, **momentum is the sole alpha signal used for ranking stocks.**

3.2 Trend and eligibility filters

A stock must satisfy a 250-day moving-average trend condition:

$$P_{i,t} > \text{SMA}_{250,i,t}. \quad (4)$$

The model additionally requires:

- at least 180 valid daily return observations;
- non-missing momentum and price data;
- annualized realized volatility between 5% and 100%;
- membership in the point-in-time eligible universe, when the eligibility matrix is available.

These conditions are risk and data-quality filters rather than additional alpha factors.

3.3 Portfolio selection and diversification

Eligible stocks are ranked in descending order of momentum score. The portfolio can contain at most ten stocks.

To reduce unnecessary turnover, an existing holding receives a rank buffer: it may remain eligible if it lies within the top

$$10 + 5 = 15 \quad (5)$$

ranked stocks, subject to the diversification constraints below. New positions must originate from the top ten ranked candidates.

Sector cap. No more than two stocks from the same sector may be selected.

Correlation constraint. Using trailing 60-day daily returns, a candidate is rejected if its maximum pairwise correlation with any already selected stock exceeds 0.65:

$$\max_{j \in \mathcal{P}} \rho_{ij} > 0.65 \quad \Rightarrow \quad i \text{ is rejected.} \quad (6)$$

The final selection is therefore not merely the ten highest momentum stocks. It is a momentum-ranked portfolio subject to explicit structural diversification.

3.4 Risk-based portfolio weights

After the stock-selection stage, trailing realized volatility is used for allocation:

$$\sigma_{i,t} = \text{Std}(r_i)\sqrt{252}, \quad (7)$$

and target weights are

$$w_i = \frac{1/\sigma_{i,t}}{\sum_{j \in \mathcal{P}} 1/\sigma_{j,t}}, \quad \sum_i w_i = 1. \quad (8)$$

This does not make volatility an alpha signal. A highly volatile stock can still be selected if it has sufficiently strong momentum, but it receives a smaller allocation relative to a lower-volatility selected stock.

3.5 Rebalancing and transaction costs

The portfolio is reviewed on the first available trading day of each calendar month. Positions no longer satisfying the target selection are exited; retained positions are trimmed or increased toward their new target weights; and new positions are initiated where necessary.

Every buy, sell, and trim incurs the required friction:

$$\text{Fee} = 0.001 \times |\text{Trade Value}|. \quad (9)$$

The implementation uses explicit cash accounting, integer share quantities, average-cost position accounting, and separate realized and unrealized P&L tracking.

Tools and Software Used

The strategy and backtest are implemented in **Python**. The principal libraries are:

- **pandas** for time-series processing and portfolio accounting;
- **NumPy** for numerical calculations;
- **Matplotlib** for performance visualization.

The submitted repository contains the complete workflow for data preparation, portfolio construction, rebalancing, transaction-cost accounting, performance evaluation, and benchmark analysis.

Results and Performance Metrics

The full in-sample backtest covers the competition modelling period from 1 January 2021 to 31 December 2025.

Table 1: Portfolio performance: 2021–2025

Category	Metric	Result
Capital	Starting Capital	Rs. 1,00,00,000.00
	Final Portfolio Value	Rs. 7,38,66,737.20
	Total Net P&L	+Rs. 6,38,66,737.20
Returns	Absolute Return	638.67%
	CAGR	49.21%
Risk	Maximum Drawdown	-27.92%
	Sharpe Ratio ($R_f = 0$)	1.84
	Annualized Volatility	23.75%
Trading	Total Rebalance Trades	576
	Realized Transaction Win Rate	77.29%
	Realized Transaction Gain/Loss Ratio	0.79
	Realized Transaction Profit Factor	2.70
	Transaction Friction Paid	Rs. 11,81,632.72

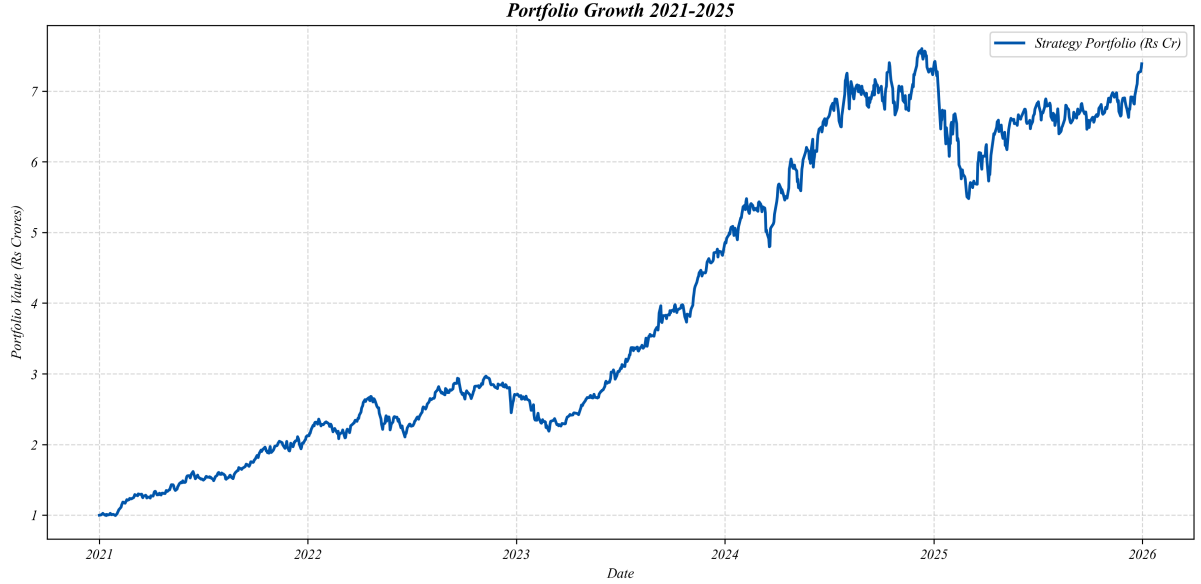


Figure 2: Portfolio value over the main 2021–2025 backtest period.

The portfolio compounded from Rs. 1 crore to approximately Rs. 7.39 crore. The resulting 49.21% CAGR is exceptionally strong and should therefore be interpreted together with the risk statistics rather than in isolation. The equity curve is not monotonic: it contains extended consolidation phases and substantial drawdowns, with the worst peak-to-trough decline reaching 27.92%. The annualized volatility of 23.75% and Sharpe ratio of 1.84 indicate that the return was not generated solely through extreme volatility.

The accounting is internally reconciled:

$$1.00 \text{ Cr} + 5.8332 \text{ Cr (realized P\&L)} + 0.5535 \text{ Cr (unrealized P\&L)} = 7.3867 \text{ Cr}, \quad (10)$$

with a reconciliation error of Rs. 0.00. Transaction costs are included directly in buy and sell accounting rather than being added as a post-hoc adjustment.

At the realized-transaction level, the gain/loss ratio is below one because average losing transactions are somewhat larger than average winning transactions. However, the high 77.29% win rate produces a profit factor of 2.70. At the completed-position level, the statistics are also positive: 65.62% of 96 completed positions were profitable, with a position-level gain/loss ratio of 1.45 and profit factor of 2.76.

Out-of-Sample Stress Test: 2026 H1

Following the challenge framing, the period from 1 January 2026 to 30 June 2026 is treated as a **forward-looking out-of-sample stress test**. The strategy rules and parameters were kept unchanged.

Table 2: Out-of-sample performance: 2026 H1

Category	Metric	Result
Capital	Starting Capital	Rs. 1,00,00,000.00
	Final Portfolio Value	Rs. 1,14,61,106.18
	Total Net P&L	+Rs. 14,61,106.18
Returns	Absolute Return	14.61%
	Annualized CAGR equivalent	32.08%
Risk	Maximum Drawdown	-10.37%
	Sharpe Ratio ($R_f = 0$)	1.30
	Annualized Volatility	23.96%
Trading	Total Rebalance Trades	47
	Realized Transaction Win Rate	54.55%
	Realized Transaction Gain/Loss Ratio	1.77
	Realized Transaction Profit Factor	2.13
	Transaction Friction Paid	Rs. 35,006.42

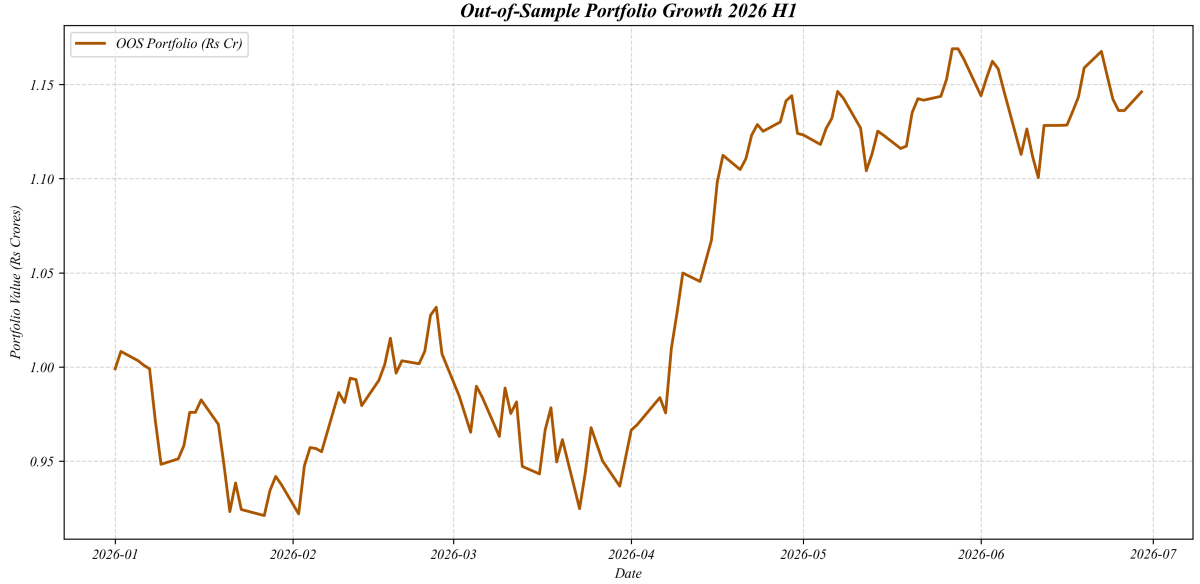


Figure 3: Out-of-sample portfolio value during the 2026 H1 stress-test window.

The OOS result is encouraging for three reasons. First, the strategy remained profitable without changing its core methodology, producing a realized six-month return of 14.61%. Second, the Sharpe ratio remained positive at 1.30, indicating that performance was not simply a consequence of a single extreme upward move. Third, the maximum drawdown was limited to 10.37% during the observed window.

Performance is naturally weaker than the exceptional 2021–2025 in-sample compounding rate, but this degradation is not unexpected. A useful robustness test should not assume that an unusually favorable historical regime repeats unchanged. The persistence of positive returns and a Sharpe ratio above one under an unchanged rule set provides evidence that the strategy is not entirely dependent on the in-sample path.

The OOS accounting also reconciles exactly:

$$1.00 \text{ Cr} + 0.05476 \text{ Cr (realized P\&L)} + 0.09135 \text{ Cr (unrealized P\&L)} = 1.14611 \text{ Cr.} \quad (11)$$

However, the OOS statistics should not be overinterpreted. The sample contains only approximately six months and only six completed position episodes. In particular, the position-level win rate of 16.67% corresponds to one winning and five losing completed positions; the result is therefore statistically too small to infer a stable long-run win probability. The portfolio-level return, drawdown, and Sharpe ratio are more informative for this short stress-test horizon.

Benchmark Comparison

The competition requires comparison against an appropriate market benchmark over the identical 2021–2025 period. Given that the strategy draws candidates from large-, mid-, and small-cap segments, **Nifty 500** is a natural broad-market benchmark because it better represents the multi-cap opportunity set than a large-cap-only index.

For the final submission, benchmark results should be generated using the same start and end dates and inserted from the repository’s benchmark backtest rather than relying on externally quoted headline returns. The comparison should include at minimum:

- absolute return and CAGR;
- maximum drawdown;
- annualized volatility and Sharpe ratio;
- relative portfolio growth against the benchmark.

This ensures an apples-to-apples comparison under identical dates and data conventions. No benchmark return is stated here without running the corresponding benchmark calculation.

Limitations and Discussion

The results are strong, but several limitations should be stated explicitly.

1. **Execution assumptions:** the simulation charges the mandated 0.1% transaction cost but does not separately model bid–ask spreads, market impact, or order-fill uncertainty.
2. **Momentum regime dependence:** momentum strategies can experience sharp reversals when market leadership changes abruptly.
3. **Concentrated portfolio:** a maximum of ten holdings improves focus but increases idiosyncratic exposure relative to a broad index.
4. **Correlation instability:** the 60-day correlation matrix is backward-looking and can change rapidly during market stress.
5. **Short OOS horizon:** six months provides useful stress-test evidence but is insufficient to establish long-run statistical certainty.
6. **Historical performance is not a forecast:** the 2021–2025 CAGR reflects the realized market path and should not be interpreted as an expected future return.

Overall, the strategy is designed around a transparent and reproducible principle rather than a collection of stock-specific discretionary choices. Its strong in-sample performance is accompanied by meaningful drawdowns and explicit trading friction, while the unchanged 2026 H1 stress test remains positive. The principal evidence in favor of the approach is therefore the combination of a simple momentum-based alpha rule, explicit diversification constraints, risk-aware allocation, internally reconciled accounting, and positive performance under an unseen forward period.

Appendix: Factor Exploration and Parameter Selection

The final strategy was obtained after an initial phase of exploratory model construction. The original formulation considered two quantities in the stock-ranking stage: intermediate-term momentum and realized volatility. The general form of the exploratory alpha score was

$$S_{i,t} = \alpha Z_{i,t}^{\text{mom}} + (1 - \alpha) Z_{i,t}^{\text{vol}}, \quad (12)$$

where $Z_{i,t}^{\text{mom}}$ and $Z_{i,t}^{\text{vol}}$ denote the cross-sectionally standardized momentum and volatility measures, respectively, and α controls their relative contribution to the ranking score.

Different choices of the relative coefficients were explored during the development stage. This exercise was useful not only for comparing numerical performance, but also for understanding the distinct roles played by momentum and volatility in the overall portfolio construction framework.

Momentum was found to provide the primary directional information for identifying stocks with persistent price strength. Volatility, on the other hand, was already incorporated elsewhere in the strategy through the portfolio allocation mechanism. Once the stocks were selected, their portfolio weights were determined using inverse-volatility weighting,

$$w_i = \frac{1/\sigma_{i,t}}{\sum_{j \in \mathcal{P}} 1/\sigma_{j,t}}. \quad (13)$$

Thus, volatility was already influencing the amount of capital allocated to each selected stock. Including it again as a component of the alpha score introduced an overlap between the *selection* and *allocation* stages of the model.

Based on these observations, the final implementation adopts a simpler separation of responsibilities:

- **Momentum** determines the ranking and hence the alpha-driven stock selection;
- **Trend, liquidity/data availability, and volatility bounds** act as eligibility and risk filters;
- **Sector and correlation constraints** control portfolio diversification;
- **Volatility** determines the relative allocation among the selected stocks through inverse-volatility weighting.

Consequently, the final alpha score used in the submitted strategy is

$$S_{i,t} = Z_{i,t}^{\text{mom}}, \quad (14)$$

corresponding to a momentum coefficient of unity and a volatility coefficient of zero in the ranking stage.

The remaining parameter values, including the momentum lookback, skip period, moving-average window, correlation threshold, sector cap, holding buffer, and volatility-estimation window, were similarly selected through exploratory trials and observation of the resulting portfolio behaviour. The purpose of this process was to arrive at a coherent and stable configuration rather than to maximize a single performance metric in isolation.

It should be emphasized that the chosen parameter set is not claimed to be uniquely optimal. There remains scope for further systematic investigation of alternative parameter combinations, rolling-window choices, and factor specifications. Such extensions could be studied using formal walk-forward validation or broader robustness analysis. For the present strategy, however, the selected configuration was preferred because it provides a clear separation between alpha generation and risk allocation while maintaining a transparent and reproducible portfolio construction process.

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